Step 1 — Installation

Goal: download Minecraft MIDI Studio and start it for the first time (about 5 minutes).

What you need

- A Windows 10 or 11 computer (64-bit).
- An internet connection at the first start (about 41 MB, only once: see below).
- **Minecraft Java Edition**, to play the music in your world. It does **not** need to be started before: the software finds the worlds of the official launcher, Modrinth, CurseForge and Prism Launcher by itself.
- About 350 MB of free disk space (program, official sounds, sample cache).

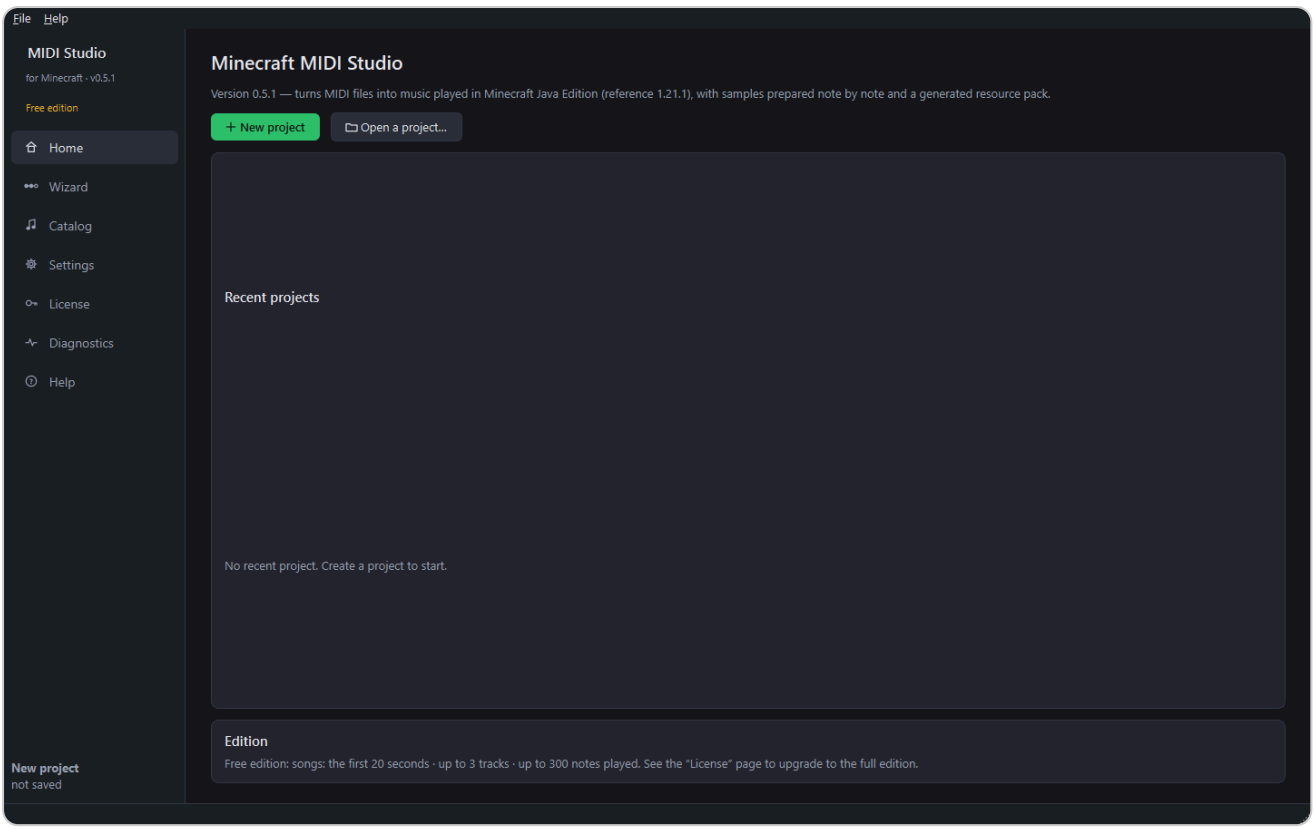
Note Block Studio is not needed. If it is installed, the full edition can also use its custom sounds.

Download

1. Open the **Download** page of the website (GitHub *Releases*).
2. Download `MinecraftMidiStudio-<version>-windows.exe`.
3. Put it in a folder of your choice, for example `Documents\Minecraft MIDI Studio`.

First start

1. Double-click `MinecraftMidiStudio.exe`.
2. Windows may show *"Windows protected your PC"* (the program is not signed with a paid certificate). Click **More info**, then **Run anyway**. This only happens the first time.
3. The first start takes a few seconds longer: the program unpacks itself once into your cache folder.
4. The **Home** page opens.



The home page

The official Minecraft sounds

The instrument and mob sounds, and the textures of the icons, belong to Mojang: they are not inside the program. The first time they are needed (opening the catalog or a project):

- if Minecraft 1.21.1 is installed on the computer, the software uses its files;
- otherwise it **downloads them itself from Mojang's official servers**, like a launcher: about 41 MB, only once, every file being checked against the fingerprint published by Mojang.

The progress is shown at the bottom of the window. Without internet, the software says so; the **Download now** button of the *Settings* page starts the download again later.

Where are my files?

What	Where
Settings and license activation	`%APPDATA%\MinecraftMidiStudio`
Downloaded official sounds	`%LOCALAPPDATA%\MinecraftMidiStudio\minecraft`

What	Where
Logs, sample cache, world backups	`%LOCALAPPDATA%\MinecraftMidiStudio`
Your projects	wherever you save them (`.mmsproj` files)

Uninstall

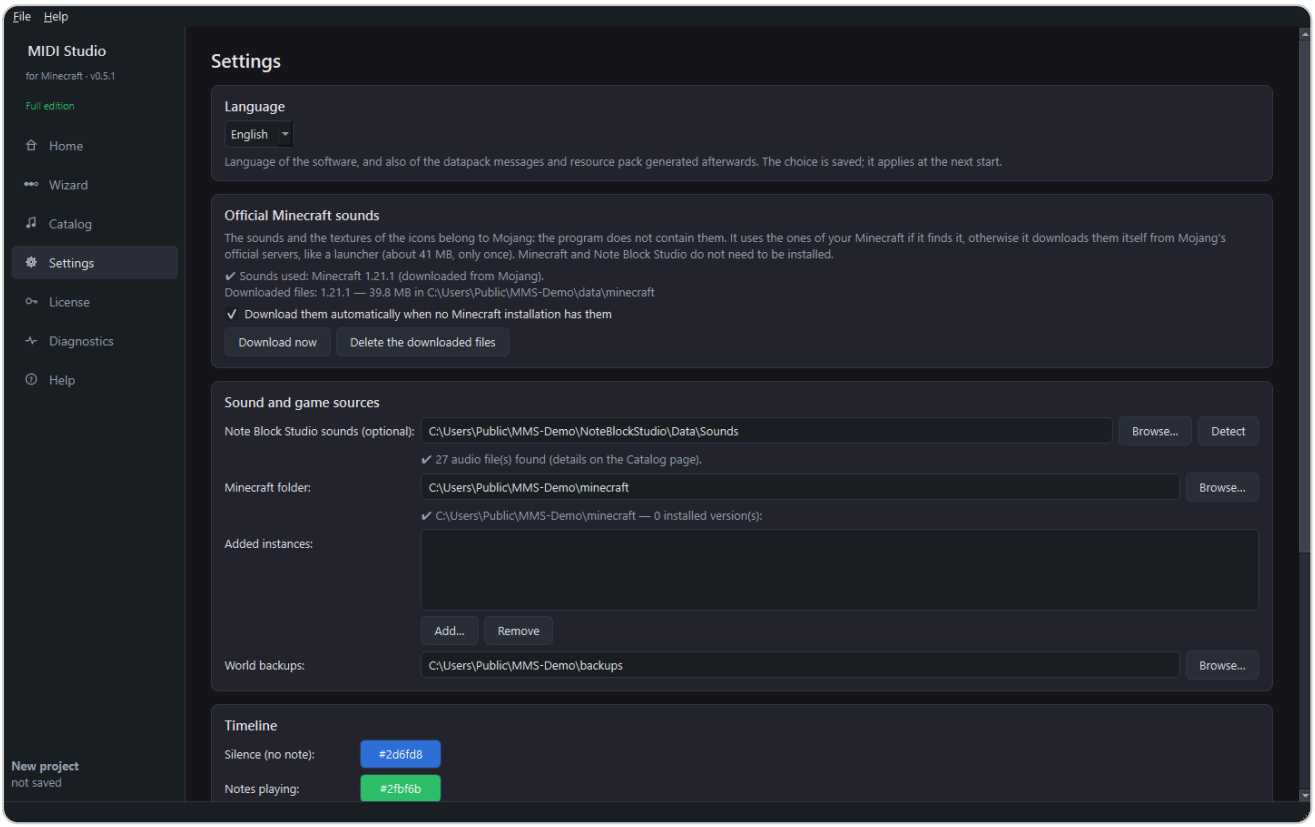
Delete `MinecraftMidiStudio.exe`. To also remove your settings, the downloaded sounds, the cache and the backups, delete the two folders above. Nothing else is installed on your computer.

Next guide: first settings.

Step 2 — First settings

Goal: choose the language and check where the sounds come from (usually nothing to change).

Open the **Settings** page from the left menu.



The settings page

Language

1. In **Language**, choose **English** or **Français**.
2. Click **Restart now** (or simply restart the program later).

The language is saved. It is also used for the messages of the datapack (in-game chat) and the description of the resource pack generated afterwards.

Official Minecraft sounds

Usually nothing to do: the software uses the sounds of your Minecraft, or downloads them itself from Mojang's official servers (see step 1). This section shows the version used and lets you:

- **Download now** — start the download again (after an internet cut, for example);
- **Delete the downloaded files** — free about 41 MB (they will be downloaded again when needed);
- untick the **automatic download** if you prefer to use only your Minecraft installation.

Sound and game sources

- **Note Block Studio sounds (optional)** — only to add its custom sounds to the full edition. Usually detected automatically (`…\Minecraft Note Block Studio\Data\Sounds`); if not, **Browse…**.
- **Minecraft folder** — leave it on *Automatic* if you use the official launcher. Modrinth, CurseForge and Prism Launcher instances are found by themselves.
- **Added instances** — only if your game folder is somewhere unusual: click **Add…** and choose the folder that contains `saves`.
- **World backups** — where the ZIP backups made before installing are stored.

Timeline and audio

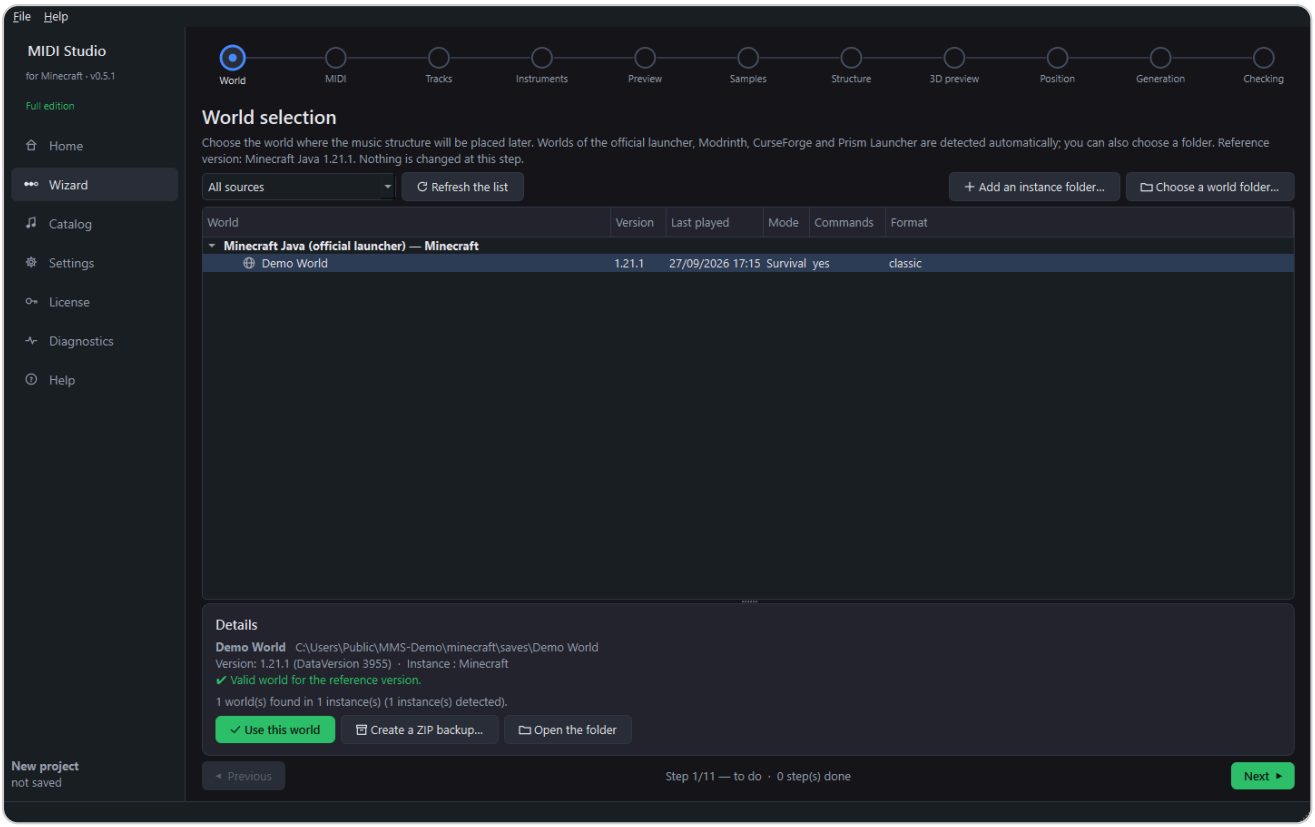
- The **colours** of the timeline (blue = silence, green = notes playing) can be changed.
- **Audio output** and **Preview volume** only affect the preview inside the software, not the game.

Next guide: choose your world.

Step 3 — Choose a world (wizard step 1)

Goal: pick the Minecraft world where the music structure will be placed.

1. On the **Home** page, click **New project**. The **Wizard** opens on step 1, *World*.
2. The list shows every world found (official launcher, Modrinth, CurseForge, Prism). A world you create while the program is open appears by itself; **Refresh the list** forces a new search.
3. Click a world: its version, game mode, commands and folder are shown below.
4. Click **Use this world**.



Choosing a world

Good to know

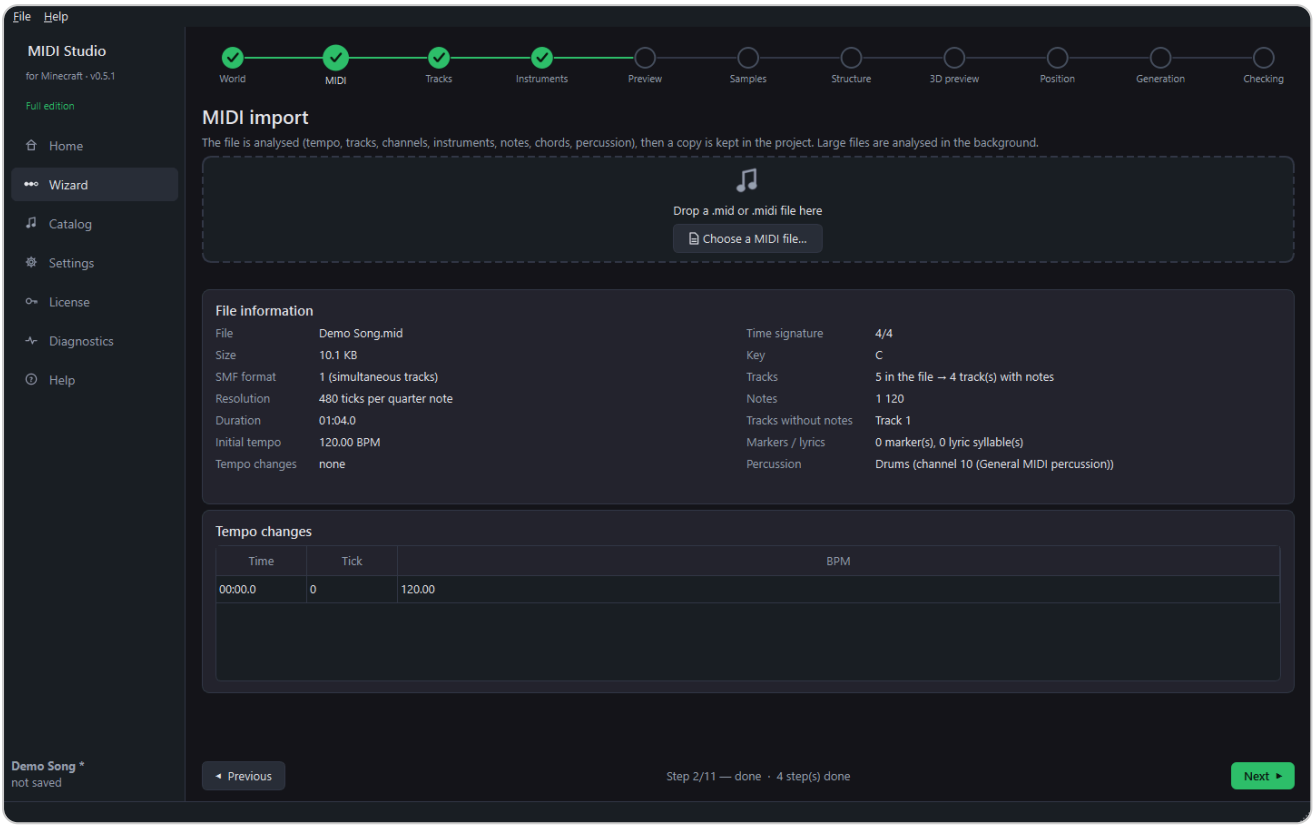
- **Commands must be allowed** in the world (the music is controlled with `/function`). If they are not: in game, *Open to LAN → Allow commands*, or use a server.
- **Close the world in Minecraft** before installing the music (step 10). The software checks it.
- **Create a ZIP backup…** copies the whole world into your backup folder without changing it. A backup is also offered automatically before any installation.
- Nothing is written into the world at this step.

Next guide: import your MIDI file.

Step 4 — Import a MIDI file (wizard step 2)

Goal: load the song you want to hear in Minecraft.

1. Click **Next** to reach step 2, *MIDI*.
2. Drag a `.mid` / `.midi` file onto the drop zone, or click **Choose a MIDI file…**.
3. The analysis runs in the background: tempo and tempo changes, duration, tracks, channels, instruments, notes, chords and percussion.



MIDI analysis

What you see

- **File information** — size, format, resolution, duration, initial tempo, time signature, key, number of tracks and notes.
- **Tempo changes** — the table on the right lists every change (they are respected in game).
- **Warnings** — for example tracks without a name, notes without an end, or a percussion track on a melodic channel.

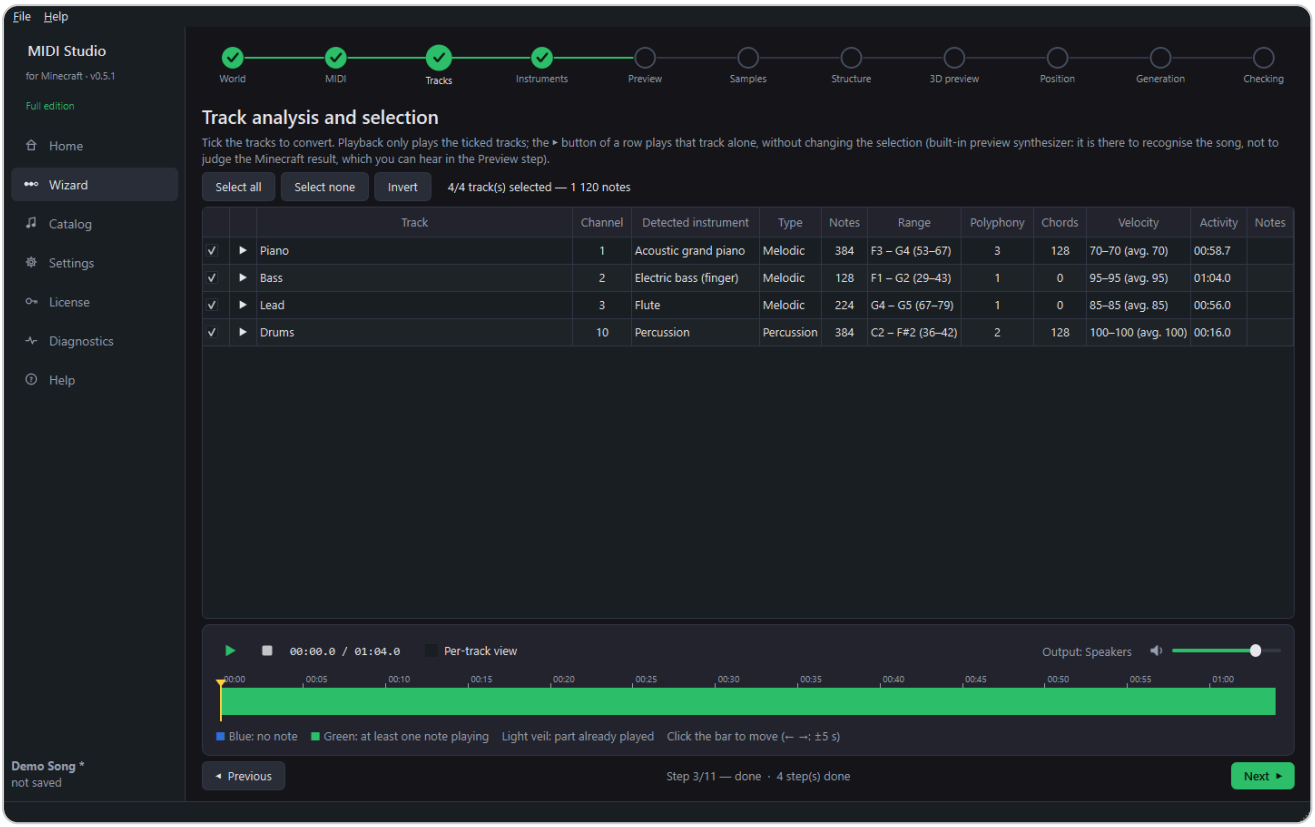
A copy of the MIDI file is stored inside the project, so the project still works if you move or delete the original file.

> **Free edition:** only the first 20 seconds of the song are played and converted. The full edition converts the whole song.

Next guide: choose the tracks.

Step 5 — Choose the tracks (wizard step 3)

Goal: keep only the tracks you want in Minecraft, and listen to them.



Track selection

1. **Tick** the tracks to convert. Buttons: **Select all**, **Select none**, **Invert**.
2. Click **Play** (or press *Space*) to hear the ticked tracks.
3. Click the ▶ button at the start of a row to hear **that track alone**, without changing the ticks. Click ■ to stop. The coloured bar then shows the notes of that track only.

The timeline

- **Blue**: no note is playing. **Green**: at least one note is playing.
- The lighter veil shows the part already played.
- Click the bar to jump anywhere; ← / → move by 5 seconds.
- **Per-track view** shows one row per ticked track.

The sound stops by itself when you go to another step.

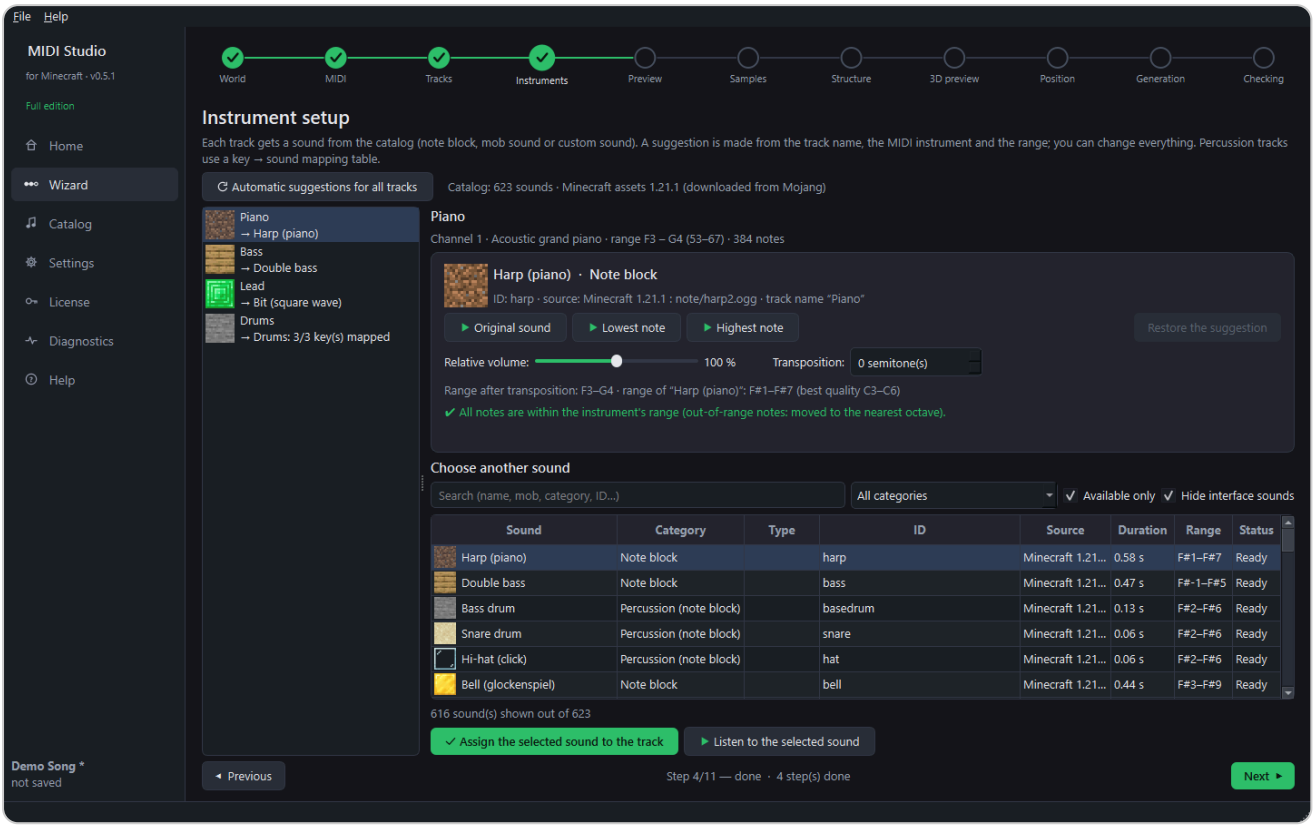
> The sound here comes from a small built-in synthesizer: it is only there to recognise the song. The real Minecraft sound is heard in step 5, *Preview*.

> **Free edition:** up to 3 ticked tracks and 300 played notes.

Next guide: instruments.

Step 6 — Instruments and drums (wizard step 4)

Goal: give each track a Minecraft sound.



Instrument setup

1. Each track gets an **automatic suggestion** (from its name, its MIDI instrument and its range). The reason is written under the sound.
2. To change it: choose another sound in the catalog on the right (search box, categories, **Listen**), then click **Assign the selected sound to the track**.
3. Adjust the **Relative volume** and the **Transposition** (in semitones) of the track.
4. **Lowest note / Highest note** let you hear the extremes of the track with the chosen sound.

Drum tracks

A percussion track uses a **key → sound** table: each drum key (kick, snare, hi-hat…) is linked to a sound and a transposition, the way Note Block Studio creators do it. Keys without a sound can be ignored or played with the snare drum. **Restore the default table** undoes your changes.

Sounds available

- **Note blocks** (official Minecraft files: those of your installation, or downloaded from Mojang).
- **Mob sounds** of the selected Minecraft version, sorted by category (voice, hurt, movement...).
- **Custom sounds** placed in the Note Block Studio `Custom` folder (full edition, if Note Block Studio is installed).

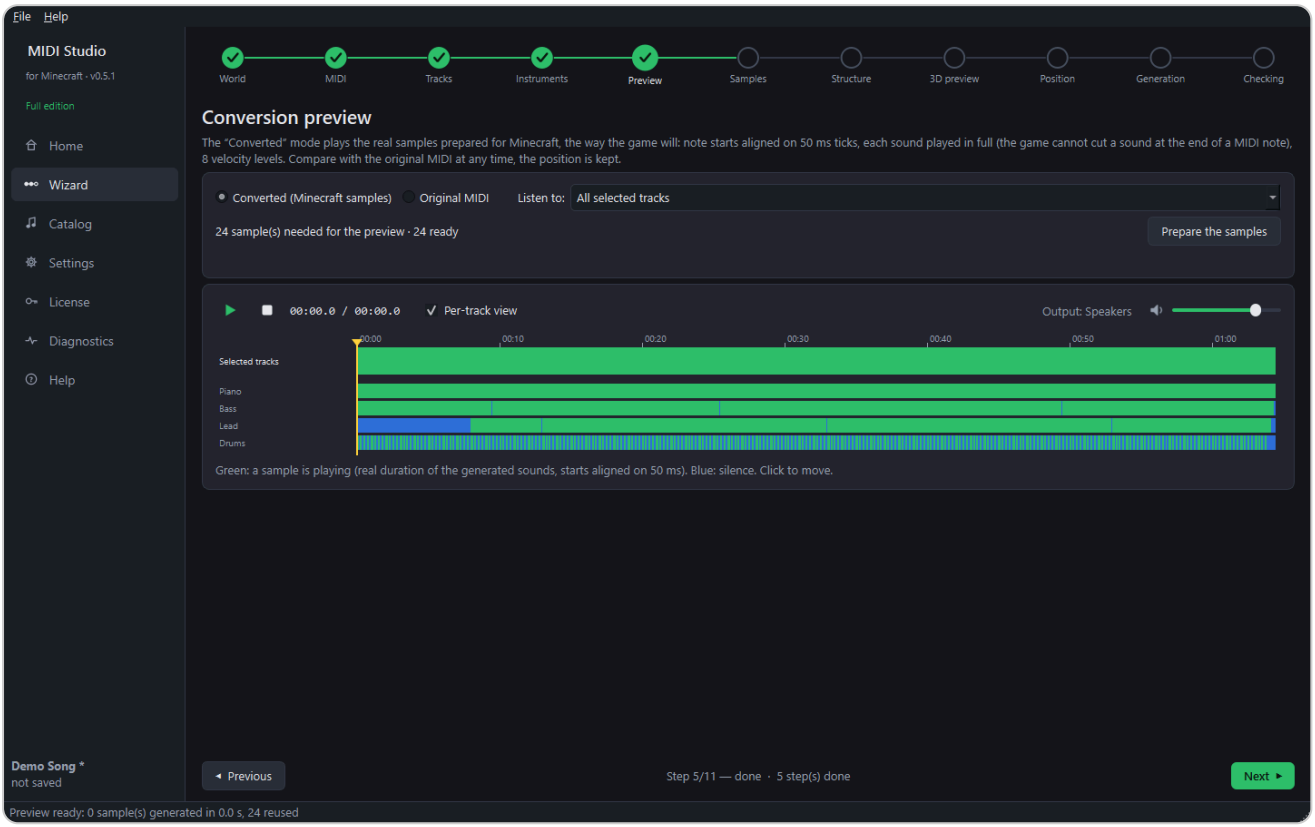
> **Free edition:** 8 basic note block instruments (no pling, bell, chime, flute, guitar, xylophones or bit), and only the voices of 3 farm animals for mob sounds.

Next guide: preview and samples.

Step 7 — Preview and samples (wizard steps 5 and 6)

Goal: hear the real Minecraft result, then prepare all the audio files.

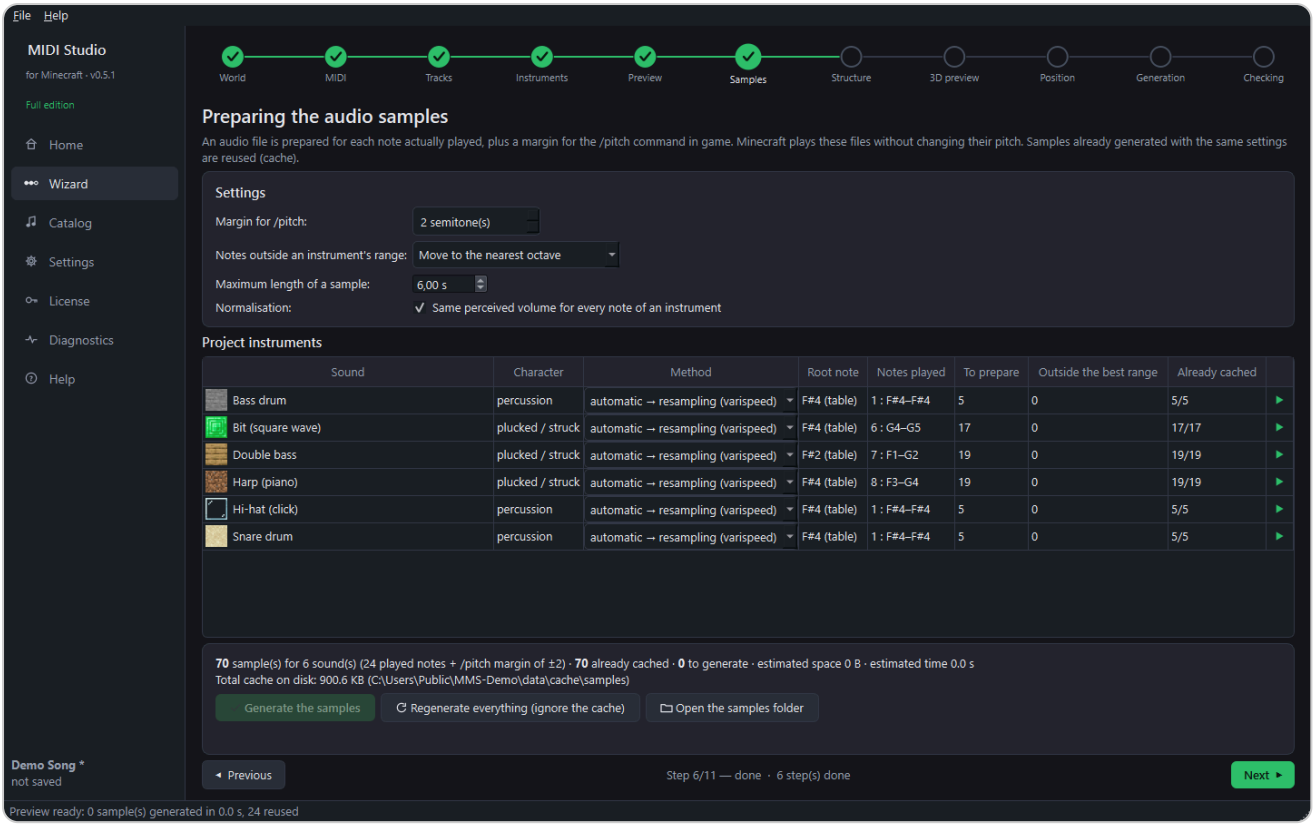
Preview (step 5)



Converted preview

1. The samples needed for the preview are prepared automatically (the first time takes a few seconds).
2. **Converted (Minecraft samples)** plays the song as the game will: notes aligned on 50 ms ticks, each sound played in full, 8 velocity levels.
3. **Original MIDI** switches back to the original at the same position, to compare.
4. **Listen to** lets you hear one track only.

Samples (step 6)



Sample generation

One audio file is prepared **for each note actually played**, so Minecraft never has to change the pitch of a sound (that is what makes the result clean).

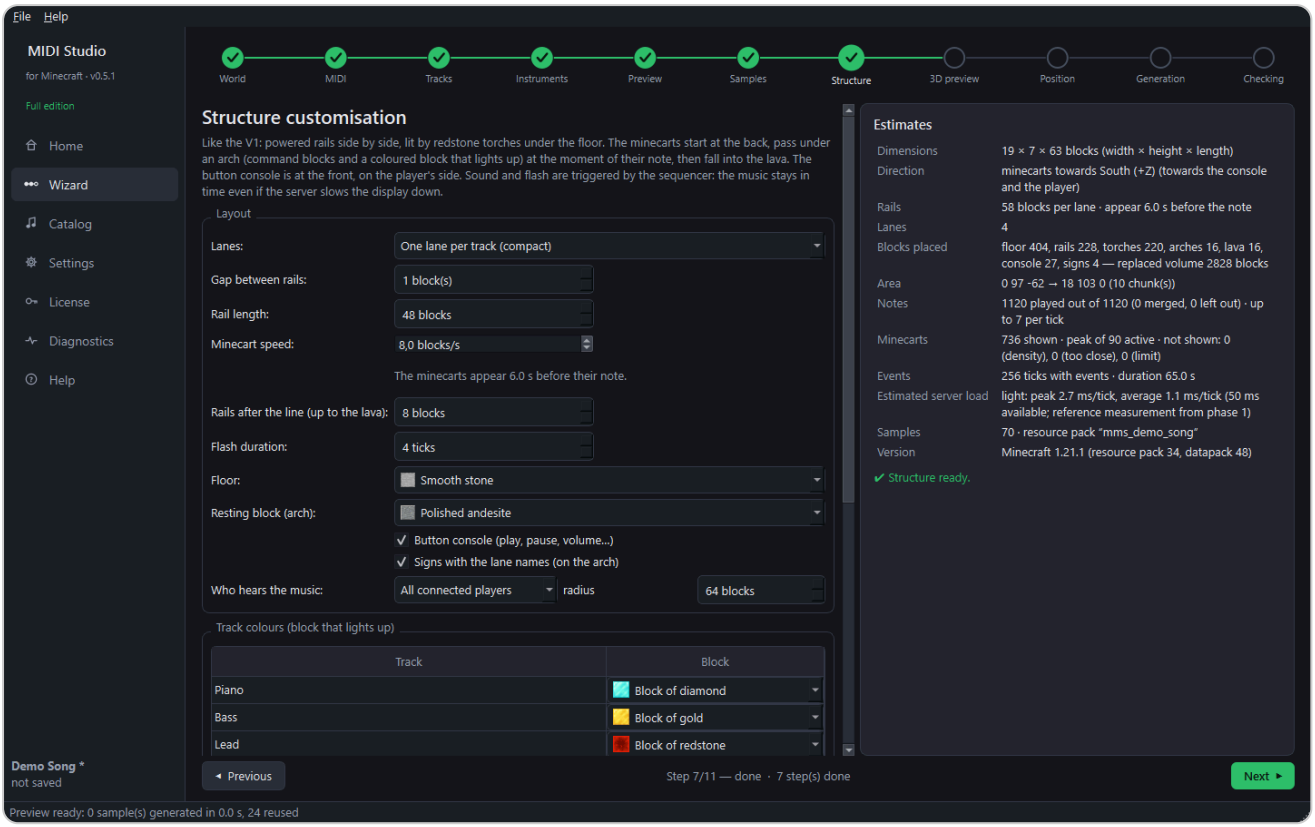
- **Margin for /pitch** — extra notes so that the in-game `/function music:pitch` command always finds a sample.
- **Notes outside an instrument's range** — moved to the nearest octave, skipped, or generated anyway.
- **Maximum length of a sample** — long sounds are shortened with a fade.
- **Normalisation** — same perceived volume for every note of an instrument.
- **Method** per instrument — *resampling* (classic note block sound) or *pitch only* (keeps the duration, better for voices).

Click **Generate the samples**. Samples already made with the same settings are reused from the cache.

Next guide: the structure.

Step 8 — The structure (wizard step 7)

Goal: choose how the music looks in the world.



Structure settings

What gets built

- **Powered rails side by side**, lit by redstone torches under the floor.
- The minecarts appear at the far end and travel towards you.
- Above the **trigger line**: an arch of command blocks and a **coloured block that lights up** when the note plays (one colour per track).
- After the line, the minecarts **fall into a lava pit**.
- At the front, on your side: a **button console** (play, pause, stop, volume −/+, tempo −/+, pitch −/+) with signs.

The sound and the flashes are triggered by the datapack's sequencer: the minecarts are only visual, so the music stays in time even if the server slows down.

Layout settings

- **Lanes** — one per track (compact), one per played note (piano roll) or one per pitch.
- **Gap between rails** — 1 = rails right next to each other.
- **Rail length** and **Minecart speed** — they decide how long before its note a minecart appears.
- **Rails after the line** — distance to the lava.
- **Floor**, **Resting block** of the arch, **Flash duration**, **Track colours**.
- **Button console**, **Signs with the lane names**, **Who hears the music** (everyone, or players near the structure).

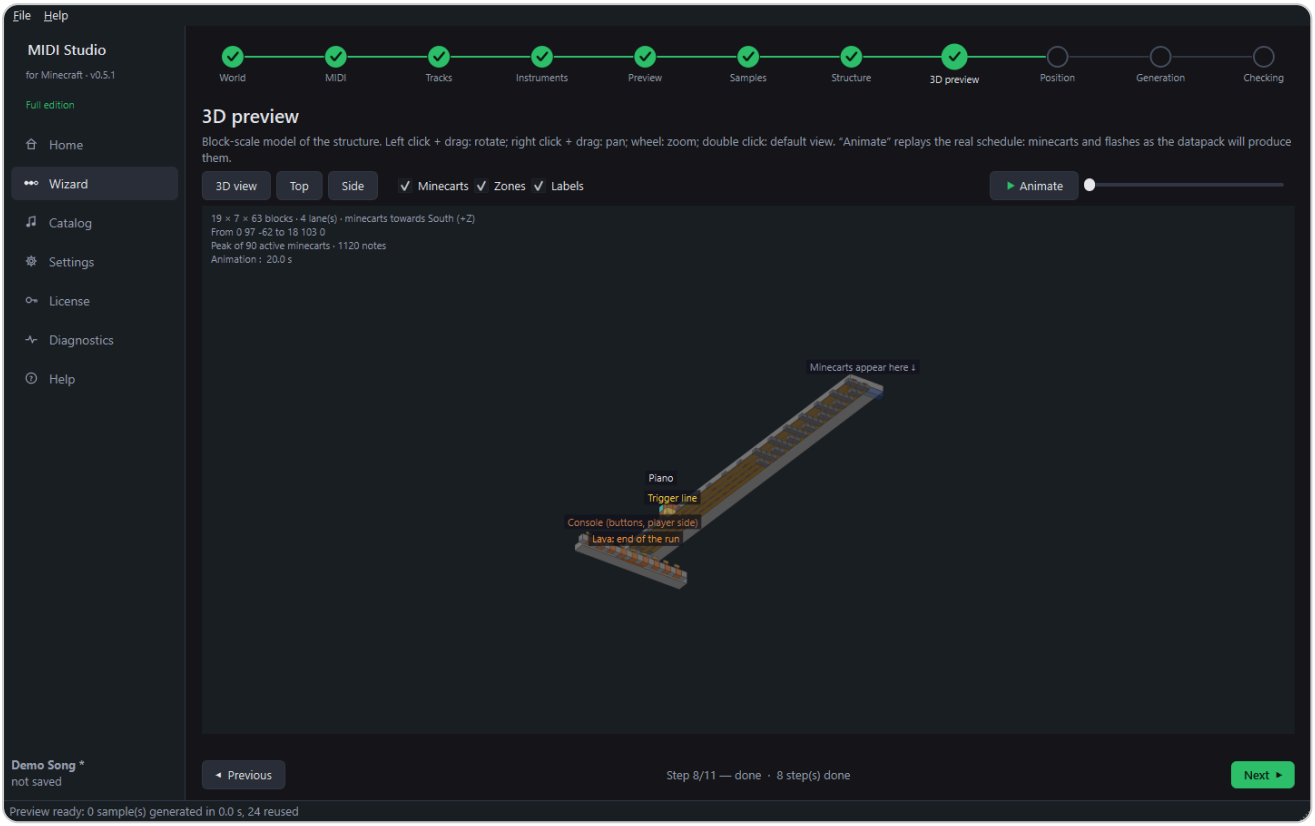
Optimisation

Choose a profile: **Best quality**, **Balanced** or **Performance**. *Custom* (full edition) unlocks every setting: maximum active minecarts, minecart density, minecart reuse, simultaneous notes per tick… The **Estimates** panel shows the size, the number of blocks, the minecarts and the estimated server load in real time.

Next guide: 3D preview and position.

Step 9 — 3D preview and position (wizard steps 8 and 9)

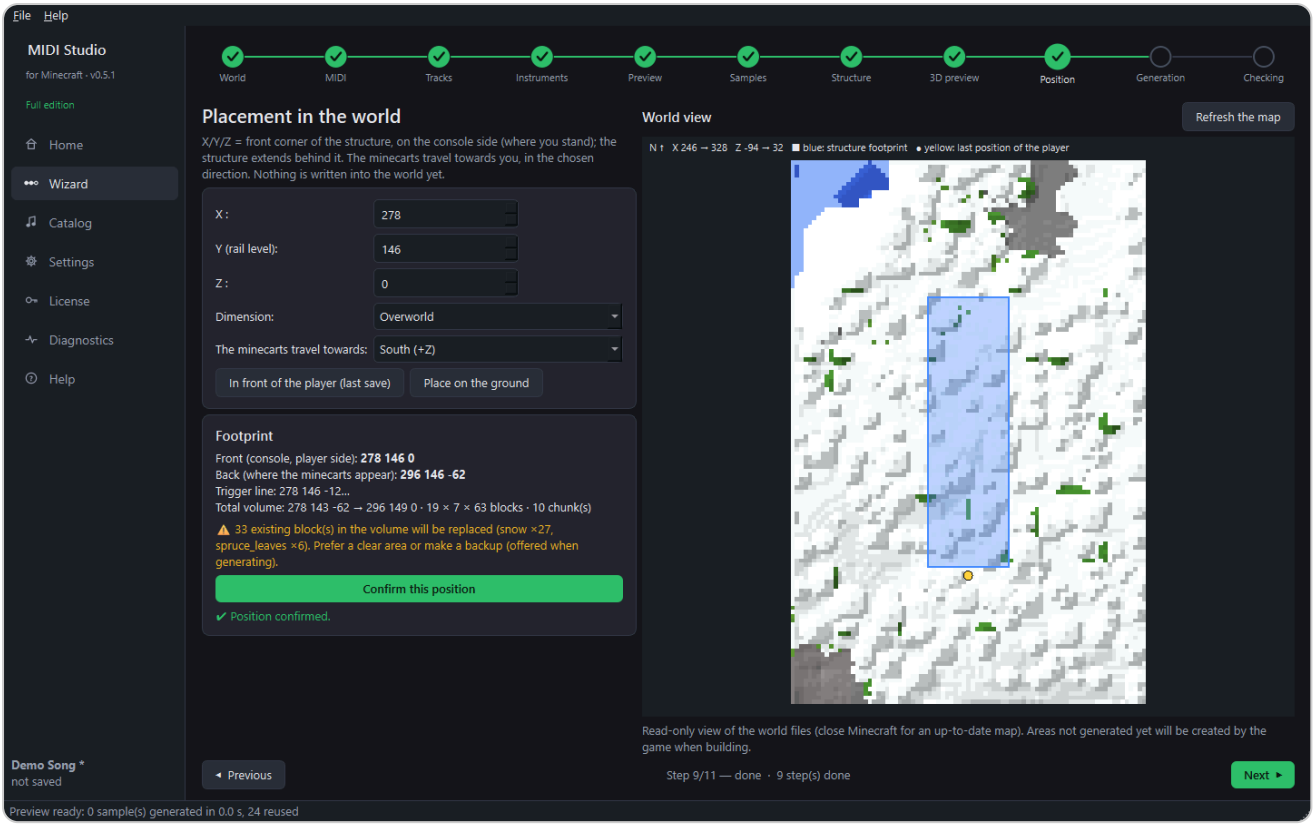
3D preview (step 8)



3D preview

- **Left click + drag**: rotate. **Right click + drag**: move. **Wheel**: zoom. **Double click**: default view.
- **3D view / Top / Side**: preset views. Layers can be hidden (minecarts, zones, labels).
- **Animate** replays the real schedule: minecarts and flashes exactly as the datapack will produce them.

Position (step 9)



Position in the world

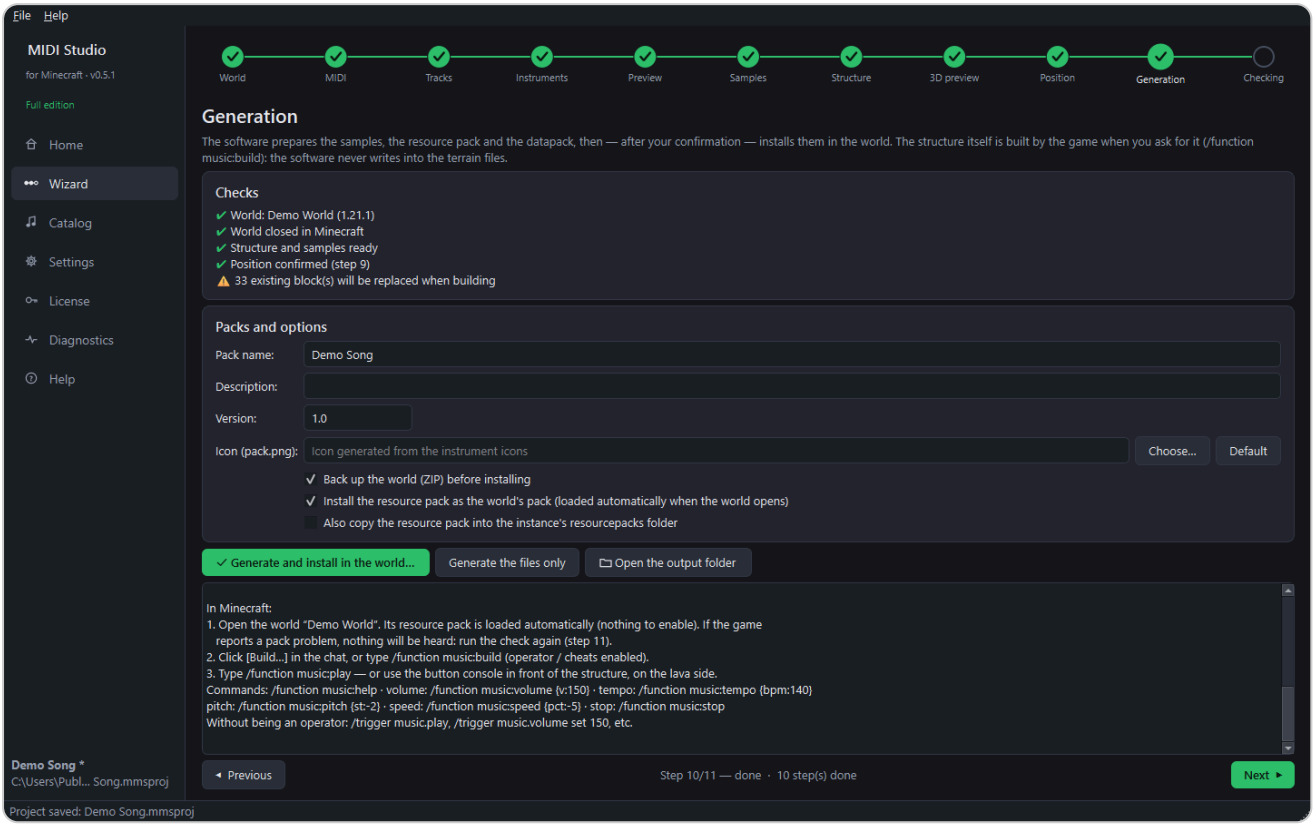
1. **X / Y / Z** is the **front corner** of the structure, on the console side, where you stand. The structure extends behind it.
2. **The minecarts travel towards** — the direction the minecarts move (towards you).
3. **In front of the player (last save)** places the console 2 blocks in front of your last saved position, centred on you.
4. **Place on the ground** sets Y just above the highest ground under the structure.
5. The **World view** map (read-only) shows the footprint in blue and your last position in yellow. The number of existing blocks that would be replaced is shown.
6. Click **Confirm this position**.

> Tip: close Minecraft before opening this step to see an up-to-date map.

Next guide: generate and install.

Step 10 — Generate, install and check (wizard steps 10 and 11)

Generation (step 10)



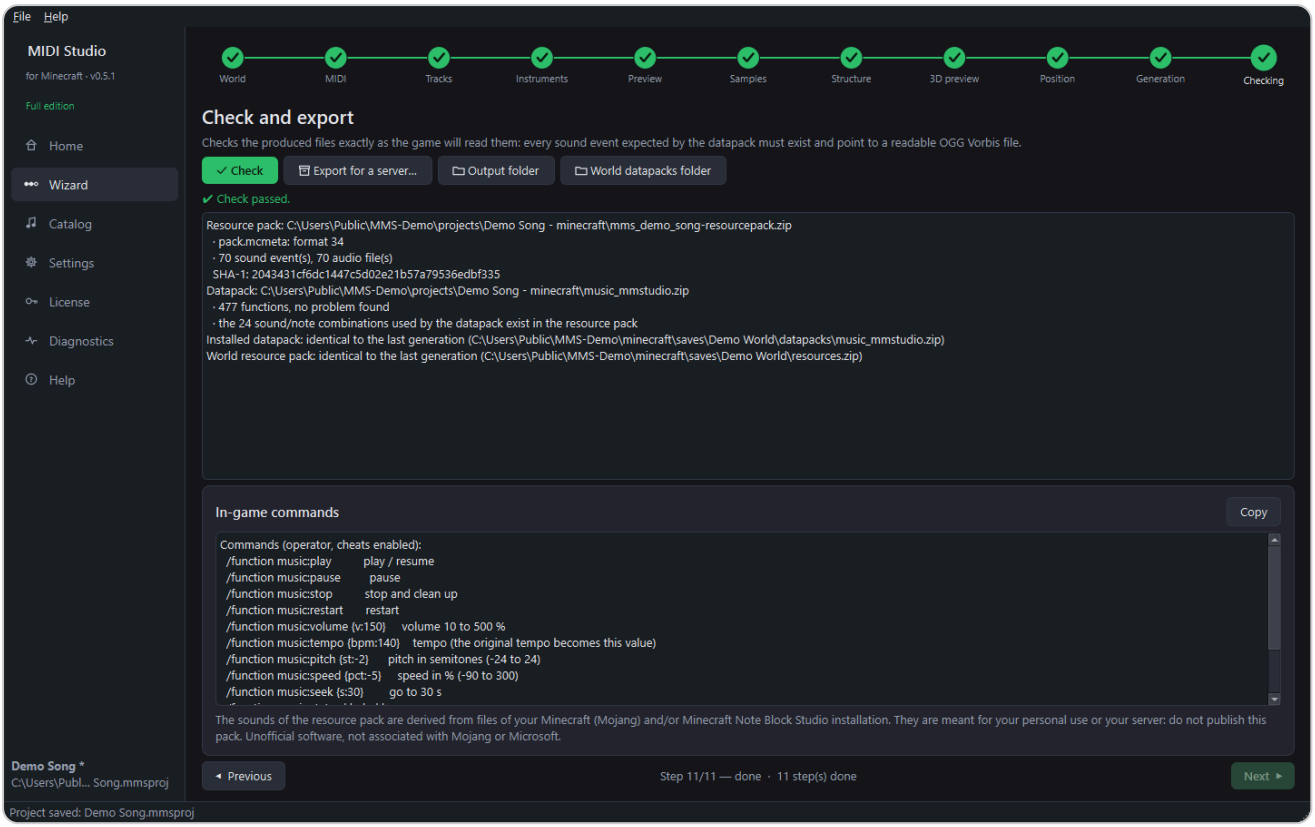
Generation

1. The **Checks** list must be green: world closed in Minecraft, structure and samples ready, position confirmed.
2. **Packs and options** — pack name, description, version, icon.
3. Click **Generate and install in the world...**.
4. A **confirmation window** lists exactly what will happen: files written, structure area, world backup. Confirm.
5. The software makes a **ZIP backup of the world**, then installs:
   - the datapack in `datapacks/music_mmstudio.zip`;
   - the **world's resource pack** (`resources.zip` in 1.21, `resourcepacks/resources.zip` in 26.x): Minecraft loads it automatically when the world opens, and only in that world.

No file is ever deleted: a replaced file is moved into the backup folder.

Generate the files only writes nothing into the world (for a server or a manual installation).

Check (step 11)



Check

Click **Check**: the files are re-read exactly as the game will read them (every sound event exists and is a valid OGG Vorbis file, every sound used by the datapack is in the resource pack, the installed files are identical). The in-game commands are listed below.

Next guide: in Minecraft.

Step 11 — In Minecraft

Goal: build the structure and play the music.

1. Start Minecraft and open the world. Its resource pack is **loaded automatically** — nothing to enable in the pack menu.
2. In the chat, click **[Build at …]**, or type:

   `/function music:build` (you must be an operator / have cheats enabled). The chunks are loaded, then the structure is built.

3. Click **[Start the music]**, press the **Play** button of the console, or type:

   `/function music:play`

All the commands

Command	Effect
`/function music:play`	play / resume
`/function music:pause`	pause
`/function music:stop`	stop and clean up
`/function music:restart`	start again
`/function music:volume {v:150}`	volume from 10 to 500 %
`/function music:tempo {bpm:140}`	tempo
`/function music:pitch {st:-2}`	pitch in semitones (−24 to 24)
`/function music:speed {pct:-5}`	speed in % (−90 to 300)
`/function music:seek {s:30}`	go to 30 seconds
`/function music:lanes`	listen to each lane (clickable list)
`/function music:status` / `help`	current settings / clickable help

Command	Effect
`/function music:remove`	remove the structure (asks for confirmation)
`/function music:uninstall`	remove the datapack's scoreboards

Players who are not operators use `/trigger music.play`, `/trigger music.pause`, `/trigger music.stop`, `/trigger music.volume set 150` (same for `tempo`, `pitch`, `speed`).

The **button console** in front of the structure does the same: play, pause, stop, volume, tempo, pitch.

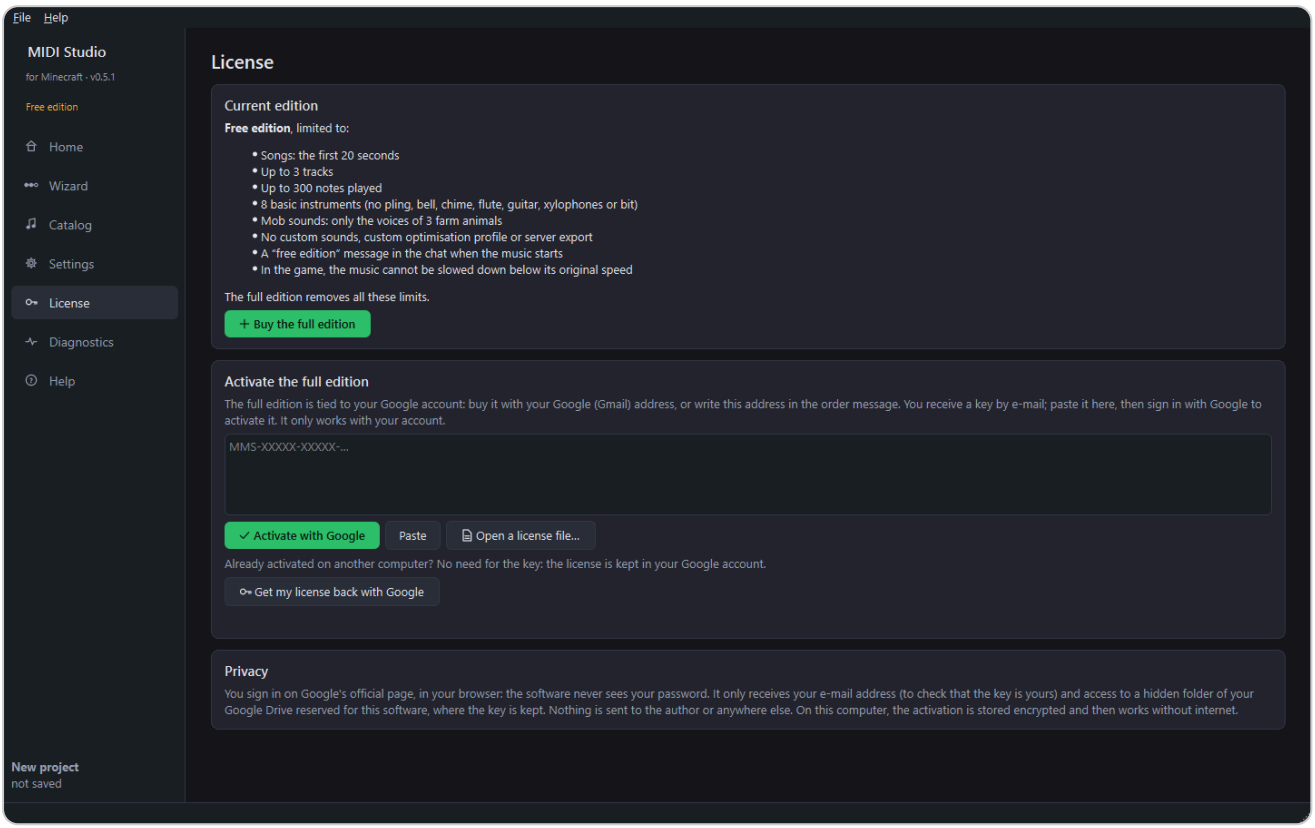
> A literal `/play music` command is impossible in vanilla Minecraft without a mod: the music is controlled with datapack functions.

Next guide: free and full editions.

Step 12 — Free and full editions, license

Free edition

Without a license the software is the **free edition**. Everything works, within these limits:



Free edition

- Songs: the first **20 seconds**.
- Up to **3 tracks** and **300 notes** played.
- **8 basic instruments** (no pling, bell, chime, flute, guitar, xylophones or bit).
- Mob sounds: only the **voices of 3 farm animals**.
- No custom sounds, custom optimisation profile or server export.
- A short *"free edition"* message in the chat when the music starts.
- In the game, the music cannot be slowed down below its original speed.

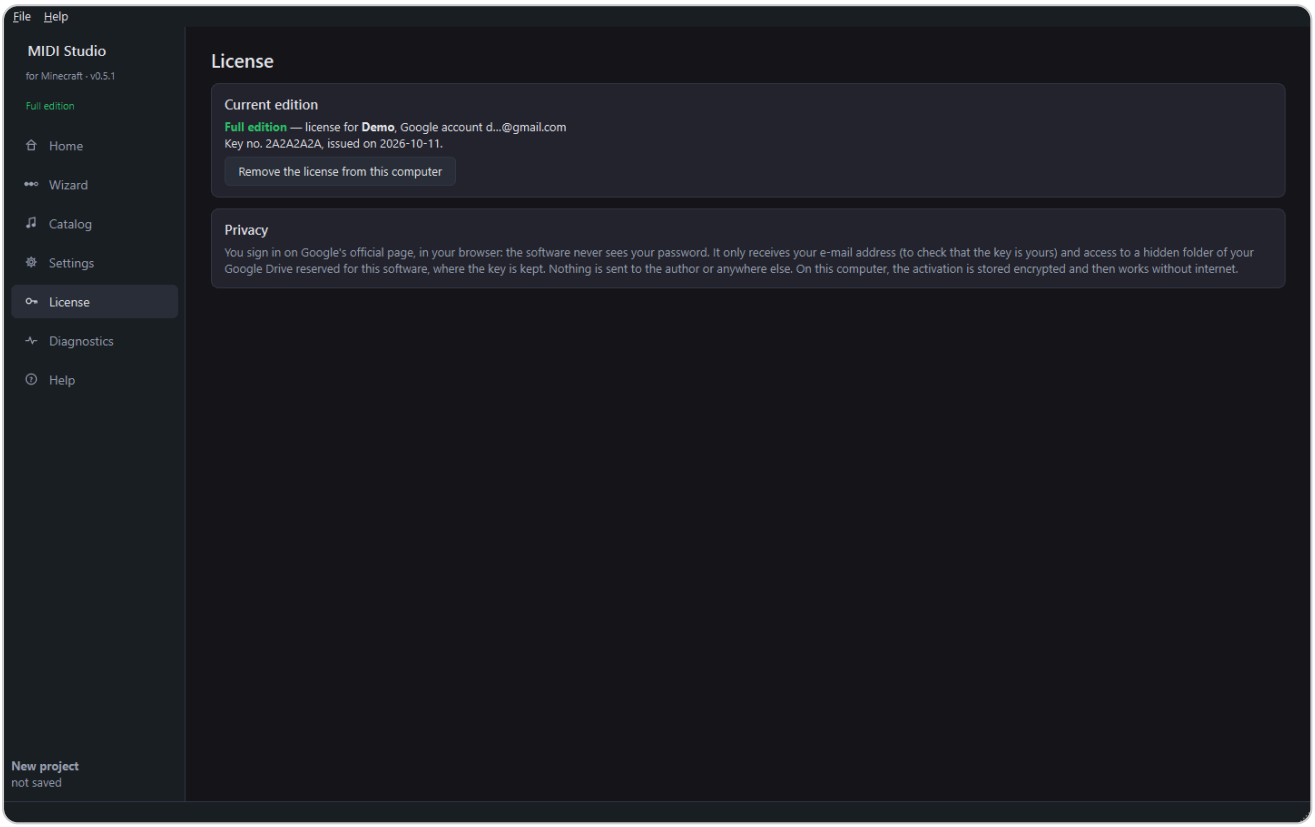
Buying the full edition

The full edition is **tied to your Google account**. A key only works with the account it was bought for: even copied, it is useless to anyone else.

1. Click **Buy the full edition** (License page): you are taken to the shop.
2. Pay with **your Google (Gmail) address**, or write it in the order message if you pay with another address.
3. You receive a **license key** by e-mail. It starts with `MMS-`.

Activating the key

1. Open the **License** page and paste the whole key.
2. Click **Activate with Google**. The software checks the key, then your browser opens **Google's official sign-in page**.
3. Sign in with the Google account shown, then accept. The page says *"Signed in"*: go back to the software.



Full edition activated

The full edition is unlocked immediately, without restarting. Afterwards the software works **without internet** on this computer.

Changing computer

The key is kept in a hidden folder of your Google Drive reserved for the software (it does not appear among your files). On another computer, on the **License** page, click **Get my license back with Google** and sign in: that's all, no key needed.

Remove the license from this computer removes the activation from this computer only; the license stays in your Google account.

Privacy

- You sign in on Google's page, in your browser: the software **never sees your password**.
- It only receives your **e-mail address** (to check that the key is yours) and access to the **software's hidden folder** in your Google Drive, where the key is kept. It cannot see your other files.
- **Nothing is sent to the author** or anywhere else. On the computer, the activation is stored encrypted.

Details: **Privacy** page of the website.

Next guide: playing on a server.

Step 13 — Playing on a server

Goal: use the music on a multiplayer server (full edition).

1. In step 11, click **Export for a server...** and choose a folder. You get:
   - `music_mmstudio.zip` (the datapack);
   - the resource pack (`.zip`);
   - `README-SERVER.txt`: the instructions, with the pack's **SHA-1**.
2. **Datapack**: copy `music_mmstudio.zip` into `<server world>/datapacks/`, then type `/reload` (or restart the server).
3. **Resource pack**: put the resource pack at a **direct download address** (HTTP or HTTPS), then in `server.properties`:

   ```
   resource-pack=<address of the file>
   resource-pack-sha1=<SHA-1 given in the instructions>
   require-resource-pack=true   (optional)
   ```

4. For the button console: `enable-command-block=true` in `server.properties`.
5. In game, as an operator: `/function music:build`, then `/function music:play`.

Players who are not operators control the music with `/trigger music.play`, `/trigger music.volume set 150`, etc.

> The sounds of the pack are Minecraft's (they belong to Mojang): use them for your server, but do not publish the pack.

Next guide: troubleshooting.

Step 14 — Troubleshooting

No sound in game

1. Did you open the world **after** installing? The world's resource pack is loaded when the world opens.
2. Watch for a pack error message when the world opens. Run **Check** (step 11) again: it must say *Check passed*.
3. Check the game's **Jukebox/Note Blocks** volume slider (*Options* → *Music & Sounds*).
4. Try `/function music:volume {v:150}`.
5. In the log of Minecraft (`.minecraft\logs\latest.log`), a line *"Unable to play unknown soundEvent"* means the resource pack is not loaded: generate and install again.

The structure is not built

- Commands must be allowed (operator / cheats).
- If the area could not be loaded in 30 s, nothing is changed: try again.

"Official Minecraft sounds not downloaded"

The first download (about 41 MB from Mojang's servers) could not be done, usually because there is no internet. Check the connection, then *Settings* → *Official Minecraft sounds* → *Download now*. An antivirus or a company firewall may also block `piston-meta.mojang.com` and `resources.download.minecraft.net`.

The license does not activate

- *"This key was bought for the Google account …"*: sign in with the Google account shown (its first letter and domain are displayed). If you paid with another address, ask the author for a key for your Google address.
- *"No license is stored in the Google account"*: use **Activate with Google** with the key received by e-mail (the first time); then **Get my license back** works on your other computers.

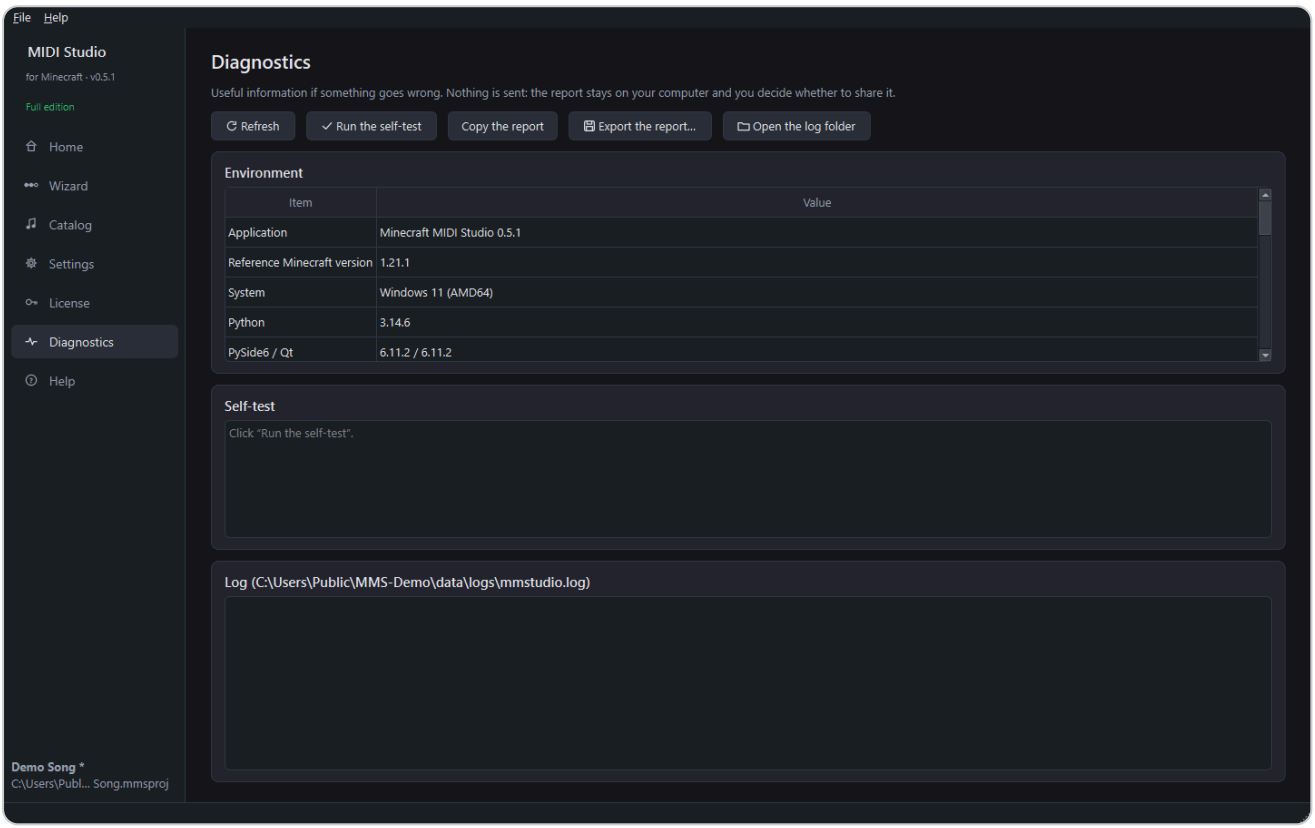
- Google's page does not come back to the software: keep the software open while signing in, and let your firewall accept the local connection (address `127.0.0.1`).
- *"This activation comes from another computer"*: normal after copying your settings; click **Get my license back with Google**.

A world does not appear in the list

- It appears by itself after a moment; **Refresh the list** forces a search.
- For an unusual game folder: **Add an instance folder…** (step 1) or *Settings → Added instances*.

The software shows an error

Open **Diagnostics**: it shows the detected environment, runs a **self-test** and shows the log. **Copy the report** or **Export the report…** to share it when you ask for help. Nothing is ever sent automatically.



Diagnostics

Undo everything in a world

- In game: `/function music:remove` (removes the structure), then `/function music:uninstall`.
- Then delete `datapacks/music_mmstudio.zip` and the world's `resources.zip` (or `resourcepacks/resources.zip`).
- Or restore the ZIP backup made before the installation.